

Ivan Kuang

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EDUCATION

University of California, Berkeley

B.S. Mechanical Engineering Undergraduate

Expected Graduation - May 2028

EXPERIENCE

Mechanical Engineering Intern | ELDAEON INC.

May 2026 - Present

- Engineered the structural chassis and internal packaging architecture for the DIONYSUS autonomous multi-sensor detection platform in Fusion 360, compressing the electronic layout by 10% through iterative DFM analysis while optimizing convective airflow to reduce peak thermal temperature by 5% under operational conditions.
- Redesigned NEMESIS unit structures through material tradeoff analysis and geometry iteration, resolving stress concentrations and improving load path efficiency while implementing DFM-driven simplifications that optimized mechanical assembly.
- Validated subsystem hardware integration and interface compatibility across both platforms, coordinating hardware-software signal endpoints to ensure reliable system functionality and cross-platform signal routing integrity.

Chassis Engineer | Cal Solar Racing

September 2024 - Present

- Designed the GEN XI Occupancy Cell in SolidWorks using a top-down CAD methodology to integrate ergonomic data, suspension hardpoints, steering geometry, and regulatory ballast within racing constraints.
- Analyzed and validated the chassis structure in ANSYS to meet WSC impact standards, correlating FEA simulations with physical so3-point bend tests to verify composite panel performance.
- Developed the manufacturing strategy for 12+ composite panels, creating custom assembly jigs and nesting patterns with GD&T specifications to minimize carbon fiber waste.

Undergraduate Student Researcher | Gu Research Group

November 2025 - Present

- Manufactured a custom Binder Jetting system, achieving a near 90% cost reduction over commercial alternatives by creating a scalable R&D platform designed to rigorously test and optimize the structural integrity of printed green bodies.
- Significantly accelerated the overall R&D cycle by developing a real-time CNN-based defect detection system utilizing image segmentation, successfully automating visual inspection protocols to reduce manual data analysis time.
- Integrated a 3-DOF mechanical camera mounting system to enable high-definition autonomous image capture, optimizing the spatial coverage and consistency of visual data used for print quality analysis.

Airframe Engineer | Space Technology and Rocketry

August 2025 - January 2026

- Designed LE3 rocket metal runners and optimized support spacing via SolidWorks FEA to ensure the structural integrity of critical fuel lines during 10 m/s impacts, a design limit derived from the terminal descent velocity calculated post-drogue parachute deployment.
- Engineered a 12-inch airframe extension to eliminate critical high-stress couplers, reducing the total chassis weight by 2% while maintaining a verified 1+ Factor of Safety under anticipated launch load simulations.

PROJECTS

Automated Matcha Dispensing System

January 2026 - May 2026

- Designed a 3-node ESP32 mechatronic system converting load cell analog voltage into calibrated mass readings via signal conditioning, enabling real-time automatic powder-to-liquid ratio calculations for precise and consistent dispensing across repeated cycles.
- Implemented ESP-NOW wireless protocol to transmit low-latency sensor telemetry between ESPs, driving a micro-gear DC motor via PWM control signals for repeatable interval-based powder dispensing.

Combat Robotics: Skofnung

August 2025 - January 2026

- Prototyped a non-traditional crank and slider shuffle drive for a 11lb combat robot, using Onshape CAD to integrate drive modules with a high-mass weapon assembly and utilizing SLA-printed 40A flexible resin feet to enhance drive torque and traction.

Power Wind Turbine

August 2024 - December 2024

- Designed and FDM 3D-printed a wind turbine system using PLA, using FEA to optimize a 178g tower that sustained a 3.5 kg static load and achieved a 500 mW peak power output through iterative blade design.

SKILLS

Relevant Coursework: SolidWorks Modeling and FEA, Solid Mechanics, Thermodynamics, Materials Behavior, Electronics for IoT, MATLAB & Python, Dynamic Systems and Feedback, Manufacturing / GD&T

Design & Simulation: SolidWorks (CAD/FEA), ANSYS, OnShape, Fusion 360, SketchUp

Programming & Data: Python, MATLAB, Jupyter Notebook, HTML

Manufacturing: 3D Printing (FDM/Binder Jetting), TIG/MIG Welding, Woodworking, CNC Machining, GD&T Design, Rapid Prototyping, Soldering